Algebra and Discrete Mathematics
ADM_A_B

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Timetable

- Lectures (prednášky)
    - Friday 9:00 – 11:50
- Tutorials (cvičenia)
    - Friday 12:00 – 13:50
- Consultations (konzultácie)
    - By appointment: xiaolu.hou@stuba.sk
    - Office 4.14, or online

Grading

- Semester work – 60 marks
  - Quiz one – 25 marks
    - Covers: Lectures 1–3, Tutorials 1–3
    - Venue: same room as for lecture
    - Week 5
  - Quiz two – 25 marks
    - Covers: Lectures 3, 4, 6, Tutorials 4–7
    - Venue: same room as for lecture
    - Week 9
  - Tutorial submissions – 10 marks
    - There will be 12 tutorial sheets
    - The best 10 submissions count
    - Each submission is worth up to 1 mark
- Final exam – 40 marks
  - To sit the final exam, you must obtain at least 30 marks during the semester.

Homework

- **Submission deadline:** Tutorials T1–T11: Wednesday, 23:59
  Tutorial T12: Thursday, 23:59
- **Submission method:** Submit via AIS or in person. If you do not have access to AIS and cannot submit in person, you may email your tutor.
- **What counts:** There will be 12 tutorial sheets; your best 10 submissions count toward your final mark.
- **Requirements:** Your solution should be clearly written, logically organised, and include the main steps of the argument, not just the final answer.
- **Marking:** Each submission is graded on a scale $0/0.5/1$:
  - 1: complete, clear, and mathematically sound solution,
  - 0.5: serious but incomplete attempt,
  - 0: no meaningful submission.
- **Late submissions:** Late submissions are not accepted.

Attendance Policy

- Attendance at tutorials is compulsory.
  - Absences must be justified by a valid medical certificate (MC).
  - In the case of more than two absences without a valid MC:
    - You must submit solutions to **ALL** questions from the missed tutorial within one week.
    - Submissions can be made via email (xiaolu.hou@stuba.sk) or in person.

Tutorials

- A summary of certain concepts from the lecture
- Discussion of tutorial questions
  - Recommendation: try to solve the questions on your own before the tutorial, then compare your work with the solutions.

Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Paths and spanning trees
- Quiz two
- Trees and networks
- Matching

Textbooks

- Andrilli, Stephen, and David Hecker. *Elementary Linear Algebra*. 5th ed. Academic Press, 2022.
  - Accessible online (free copy)
  - Alternative download link
- Anton, Howard, and Chris Rorres. *Elementary Linear Algebra: Applications Version*. John Wiley & Sons, 2013.
  - Accessible online (free copy)
  - Alternative download link
- Saoub, K. R. *A Tour Through Graph Theory*. Chapman and Hall/CRC, 2017.
  - Accessible online (free copy)
  - Alternative download link

Course materials

https://xiaoluhou.github.io/Teaching_material/

or

https://github.com/XIAOLUHOU/Teaching_material

Remarks

- **Proofs:** Included for reference and deeper understanding.
- **Errors:** Please report any mistakes by email.